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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations
May – June 2025

22MCA211 – DIGITAL IMAGE PROCESSING AND PATTERN RECOGNITION

Duration: 3 Hours

Max. Marks: 100

Note: Answer any Five full questions choosing One full question from each Unit.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Explain digital image processing and its significance. Also identify five fields where digital image processing is applied with suitable examples.	10	L*2	1	1
	b) Illustrate the elements of visual perception.	10	L2	1	1
2.	a) Explain the fundamental steps involved in digital image processing with a neat block diagram.	10	L2	1	1
	b) Explain the process of image sensing and acquisition.	10	L2	1	1
Unit – II					
3.	a) Explain histogram processing and equalization.	10	L2	2	1
	b) Explain image smoothing and image sharpening using frequency domain filters.	10	L2	2	1
4.	a) Explain the process of obtaining DFT from the continuous transform of a sampled function.	10	L2	2	2
	b) Explain the algorithm for frequency domain filtering with a block diagram.	10	L2	2	2
Unit – III					
5.	a) Explain the general image compression model with a diagram.	10	L2	3	1
	b) Illustrate gradient operators in image processing. Also provide a detailed discussion on the Roberts, Prewitt, and Sobel gradient operators.	10	L2	3	2
6.	a) Explain the model of the image degradation/restoration process.	10	L2	3	1
	b) Explain erosion, dilation with an example each.	10	L2	3	1
Unit – IV					
7.	a) Explain the different distance functions and similarity measures.	10	L2	4	2
	b) Explain the working of K-means algorithm with advantages and disadvantages.	10	L2	4	2
8.	a) Illustrate Pattern recognition. Compare and contrast classification and clustering.	10	L2	4	1
	b) Explain Bayesian belief network and conditional independence with example.	10	L2	4	1
Unit – V					
9.	a) Explain K-Nearest Neighbor algorithm with an example.	10	L2	5	2
	b) Illustrate Single layer and Multilayer perceptron. Explain the key characteristics MLP.	10	L2	5	1
10.	a) Explain the following:	10	L2	5	1
	i) Training set and test sets				
	ii) Standardization and normalization				
	b) Illustrate minimum distance classifier with an example.	10	L2	5	1

BT* Bloom's Taxonomy; L* Level; CO* Course Outcome; PO* Program Outcome.
